

TrES-3b

R-2679

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130607 · created 2026-08-30T23:29:10+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456450.83636 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.018
Transit depth (Rp/Rs)^2	2.46 +/- 0.58 [%]
Orbital Inclination (inc)	86.89 +/- 1.03
Airmass coefficient 1 (a1)	1.778 +/- 0.02
Airmass coefficient 2 (a2)	-0.55 +/- 0.15
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	None
Optimal Aperture	4.11
Optimal Annulus	8.48
Transit Duration (day)	0.08 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.018 vs 0.16309 (deviation 0.34 σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 8.05 σ)
Mid-time O-C	-3.3 min
Depth SNR	8.7
Residual scatter	1.15 %

Light curve

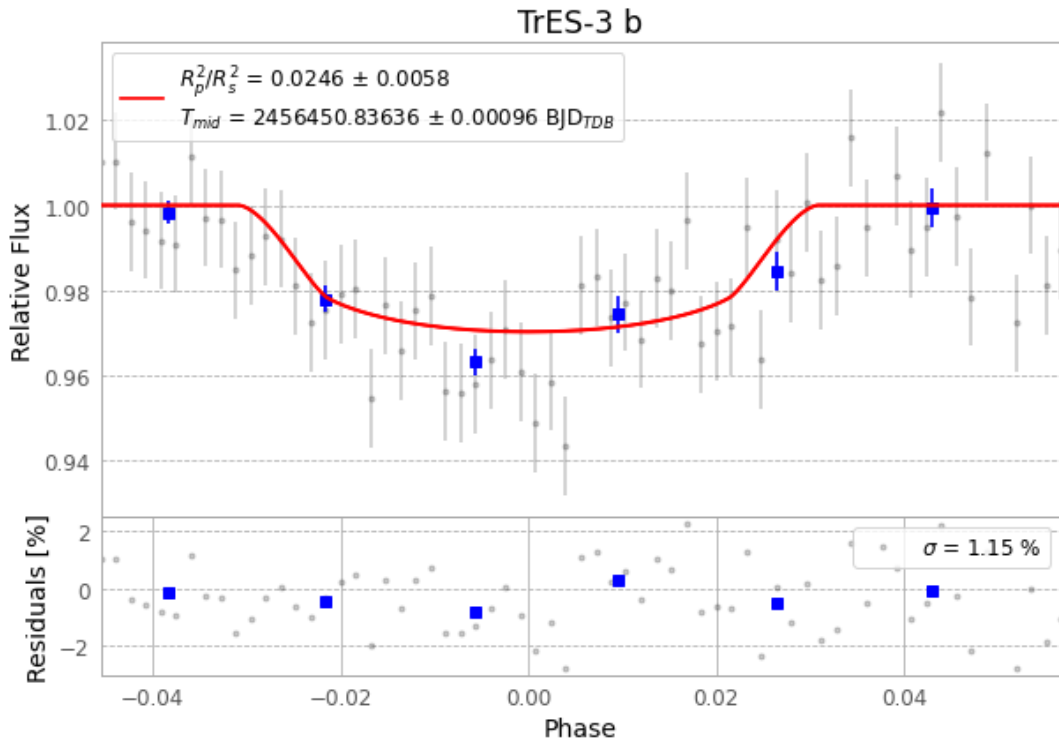


Figure 1 — FinalLightCurve_TrES-3 b_2013-06-06.png (SHA-256 d371fcf58d4a79c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a40a11437bd83b38...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3130607063312.FITS → TRES-3130607094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130607.log (83a24ba3dcdb...), FinalLightCurve_TrES-3 b_2013-06-06.png (d371fcf58d4a...), FinalLightCurve_TrES-3 b_2013-06-06.csv (82db0e608711...)

Compute resources

Wall time 190.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:29Z.